TrajScale: Decoupled Mixed-Resolution Sampling for Efficient High-Resolution Video Generation

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Abstract

High-resolution video diffusion is costly because every denoising step processes long spatiotemporal latent sequences at the target resolution. Since early steps mainly determine global structure and motion while later steps recover details, mixed-resolution sampling can reduce computation; however, resolution transition must address residual endpoint error, non-scale-equivariant VAE latents, and target-resolution re-noising. We propose **TrajScale**, a lightweight latent-space framework that performs most denoising at low resolution and decomposes the transition into three components: a LoRA-based **Endpoint Alignment Adapter (EAA)** that calibrates the residual endpoint gap; a **Cross-Resolution Latent Upsampler (CRLU)** that maps clean latents across resolutions; and **Target-Resolution Re-noising (TRR)** to re-noise the upsampled latent for continued high-resolution denoising. Operating under a model-endogenous self-distillation paradigm, EAA and CRLU are trained from supervision generated by the frozen target model, requiring no external paired videos or additional teacher. On 50-step Wan2.1 with a $368 \times 640 \rightarrow 720 \times 1248$ transition, TrajScale-Q and TrajScale-E achieve $1.83\times$ and $2.22\times$ end-to-end speedups while retaining 97.76% and 97.53% of Native-HR VBench-5. Compared to trilinear interpolation, CRLU reduces latent L_1 error by 48.6% and LPIPS by 73.3% for cross-resolution upsampling, while EAA reduces endpoint L_1 error by 25.8% and 21.0% at steps 40 and 45. TrajScale consistently exhibits a robust quality-efficiency Pareto frontier across both standard 50-step and 4-step distilled samplers.

1 Introduction

Recent latent video diffusion models such as Wan generate semantically coherent videos with realistic motion and fine details (Wan et al. 2025), but their inference cost grows aggressively with spatial resolution, video length, and sampling steps. In DiT-based models, every denoising iteration processes the full spatiotemporal token sequence. Executing the entire trajectory at the target resolution forces early steps—which mainly establish coarse structure and macro-motion—to bear the full burden of high-resolution computation, severely limiting real-time deployment. Since diffusion sampling inherently distributes generation across stages (global layout in early steps, high-frequency textures in later steps), a promising paradigm is mixed-resolution inference: modeling global content in a low-resolution latent space first, then switching to the target resolution for a short refinement suffix. This spatial-scaling approach (ModelTC 2026; Zheng et al. 2026) is fundamentally orthogonal and complementary to temporal step-distillation methods.

Yet, switching resolution within a diffusion trajectory is substantially more complex than resizing a latent tensor. The transition is governed by three coupled error sources: the residual gap to the converged low-resolution endpoint, the discrepancy between clean latent spaces at different resolutions, and the residual error left for high-resolution refinement. Directly regressing from the handoff state to a high-resolution endpoint forces a single Joint Trajectory–Scale Mapper (JTSM) to absorb timestep-dependent trajectory error, VAE-specific scale mismatch, and underdetermined high-frequency content. Our transition-step scans show that this entanglement attenuates fine detail, motivating a decomposition into localized subproblems.

We therefore propose **TrajScale**, a decoupled mixed-resolution sampling framework with three components activated at the resolution transition: a LoRA-based **Endpoint Alignment Adapter (EAA)** that calibrates the current clean prediction toward the converged low-resolution endpoint; a **Cross-Resolution Latent Upsampler (CRLU)** that maps clean latents across spatial scales; and **Target-Resolution Re-noising (TRR)**, a parameter-free operation that reconstructs a schedule-consistent high-resolution state for suffix refinement. EAA and CRLU are trained separately through model-endogenous self-distillation using supervision generated by the frozen target generator.

Our primary contributions are summarized as follows:

- **We propose TrajScale**, a lightweight mixed-resolution sampling framework that executes a low-resolution prefix followed by a short high-resolution suffix, yielding controllable quality–efficiency operating points (TrajScale-Q/E).
- **We decouple the resolution transition** into endpoint alignment, cross-resolution clean-latent upsampling, and target-resolution re-noising through EAA, CRLU, and TRR. The decomposition isolates trajectory, scale, and refinement errors that are entangled in joint mapping.
- **We develop model-endogenous dual supervision** without external paired videos or auxiliary teachers, and establish cached-prefix consistency for handoff-local adap-

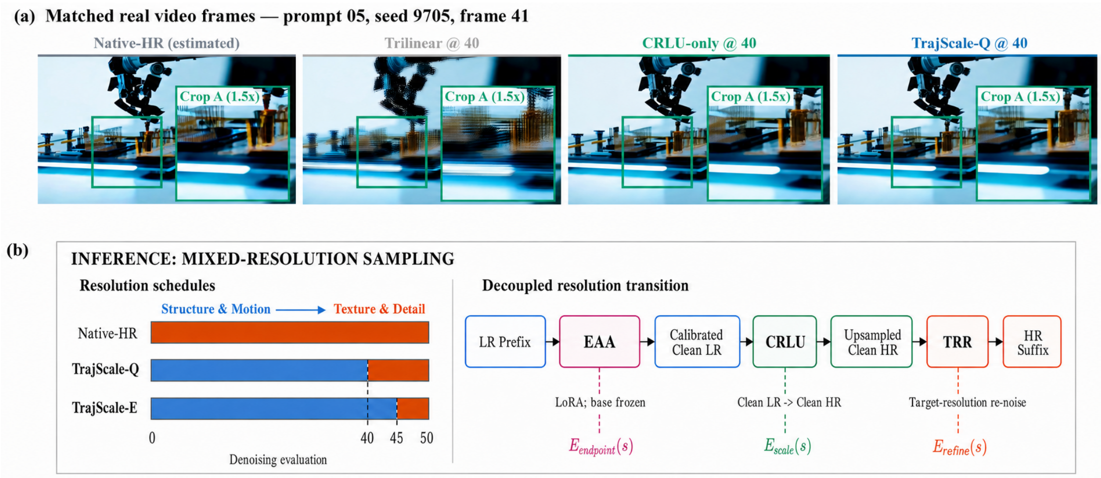

Figure 1: Real-frame comparison and TrajScale inference. (a) Content-aligned Native-HR (estimated), Trilinear, CRLU-only, and TrajScale-Q outputs for prompt 05, seed 9705, frame 41, with identical $1.5\times$ crops. The reference follows a complete 368p trajectory; transition outputs are 720p. (b) Native-HR performs all 50 evaluations at high resolution, whereas TrajScale-Q and TrajScale-E execute the first 40 and 45 evaluations at low resolution and retain 10 and 5 high-resolution suffix evaluations.

tation. Experiments on standard and distilled Wan samplers validate spatial acceleration and compatibility with few-step sampling.

2 Related Work

2.1 Latent Video Generation and Cascaded Upscaling

Latent Diffusion Models compress the generative process into a compact space for scalable high-resolution video synthesis (Rombach et al. 2022). Traditional frameworks or recent systems like LTX-2 and LUVE typically employ cascaded pipelines, using independent spatial latent upsamplers or post-generation refinement modules to elevate resolution (Blattmann et al. 2023; HaCohen et al. 2026; Zhao et al. 2026). Similarly, video super-resolution methods focus on restoring already decoded RGB outputs (Zhou et al. 2024; Xu et al. 2024).

Unlike approaches that treat upscaling as an isolated post-processing stage, TrajScale inserts CRLU directly into the frozen Wan sampling trajectory. TRR then reconstructs the target-resolution noisy state at the designated schedule level, allowing the remaining Wan steps to synthesize high-frequency content with the base model's native generative prior.

2.2 Mixed-Resolution Diffusion Inference

The changing-resolution inference in LightX2V utilizes fixed interpolation to adjust latent sizes mid-sampling (ModelTC 2026). Concurrently, RALU demonstrates that training-free latent upscaling suffers from high-frequency aliasing and a resolution-dependent noise–timestep mismatch, designing a mixed-resolution scheme for Diffusion Transformers (Jeong et al. 2025). For image flow matching, MrFlow combines low-resolution structure generation with low-noise high-resolution refinement (Zheng et al. 2026). TrajScale extends this paradigm to video by learning cross-resolution clean-latent upsampling inside the original sampling process while explicitly calibrating the residual low-resolution endpoint error before the resolution transition.

2.3 Few-Step Distillation and On-Policy Trajectory Learning

To accelerate inference, temporal distillation methods minimize sampling steps via trajectory regression or consistency mapping (Luo et al. 2023; Yin et al. 2024; Zhai et al. 2024). Recent works like Self-Forcing and AnyFlow further emphasize enforcing strict consistency between student training states and actual inference trajectories to mitigate distribution shift (Huang et al. 2025; Gu et al. 2026).

TrajScale focuses on spatial rather than temporal computation, making it orthogonal to step reduction. Since EAA remains inactive prior to the transition step, its parameter updates do not alter the low-resolution prefix trajectory. Leveraging this local activation property, Section 3.2 establishes the consistency conditions that allow cached prefix states to be directly accessed during inference without training–inference distribution drift.

3 Method

3.1 Decoupled Mixed-Resolution Transition

Let $v_\phi(x_k, k, c)$ denote a frozen Wan flow predictor, where x_k is the latent state before the k-th denoising evaluation, c is the text condition, and T is the total number of evaluations. Superscripts L and H denote the low- and target-resolution latent spaces, respectively. For a transition after evaluation s, x_s^L is the low-resolution state presented to the transition operator. We use z_T^L to denote the endpoint obtained by completing the native low-resolution trajectory under a fixed prompt, random seed, initial noise, scheduler, and guidance configuration. We further denote by (z_0^L, z_0^H) a pair of clean Wan-VAE latents encoded from low- and high-resolution versions of the same video.

A direct resolution transition could be learned with a single Joint Trajectory–Scale Mapper (JTSM),

$$M_\omega : (x_s^L, s, c) \mapsto z_0^H.$$

Such a mapper, however, must simultaneously remove the residual error between the transition-time prediction and the converged low-resolution endpoint, translate between two VAE-specific latent distributions, and infer high-frequency content that is not uniquely determined by the low-resolution input. These responsibilities depend on different variables: the first varies strongly with the transition step and scheduler, the second is primarily a cross-scale representation problem, and the third is better resolved by the pretrained generator's high-resolution prior. Learning them through one regression target therefore entangles trajectory completion, resolution conversion, and detail synthesis.

TrajScale instead factorizes the transition into three domain-specific operations:

$$x_s^L \xrightarrow{\text{EAA}:v_{\phi+\Delta\theta_s}} \tilde{z}_s^L \xrightarrow{U_\psi^{\text{CRLU}}} \tilde{z}_s^H \xrightarrow{\text{TRR}} x_{s+1}^H.$$

The Endpoint Alignment Adapter (EAA) corrects only the residual discrepancy to the converged low-resolution endpoint. The Cross-Resolution Latent Upsampler (CRLU) then maps the calibrated clean latent to the target spatial scale. Finally, Target-Resolution Re-noising (TRR) reconstructs a scheduler-compatible high-resolution state from which the frozen model continues its remaining denoising evaluations. EAA and CRLU are trained independently, while TRR is parameter free.

This factorization yields a design-oriented decomposition of transition error. Let $E_{\text{endpoint}}(s)$ represent the residual low-resolution endpoint discrepancy, $E_{\text{scale}}(s)$ the cross-resolution clean-latent reconstruction error, and $E_{\text{refine}}(s)$ the error left to the high-resolution suffix. The transition can be viewed conceptually as

$$E_{\text{transition}}(s) \approx E_{\text{endpoint}}(s) + E_{\text{scale}}(s) + E_{\text{refine}}(s).$$

A later transition generally reduces the first term and moves the CRLU input closer to its clean-latent training distribution, but it also shortens the high-resolution suffix available to synthesize fine detail. The transition step therefore controls a quality–efficiency trade-off, whereas the three TrajScale operations isolate the error sources that a monolithic JTSM must absorb jointly.

3.2 Model-Endogenous Trajectory–Scale Learning

TrajScale learns trajectory-side calibration and scale-side reconstruction from two complementary supervision streams generated entirely by the frozen target model. It requires neither external paired videos, pretrained super-resolution weights, nor an auxiliary teacher. The two streams share the same generator and VAE but correspond to different conditional distributions:

$$\underbrace{(x_s^L, z_T^L)}_{\text{trajectory-side supervision}} \quad \text{and} \quad \underbrace{(z_0^L, z_0^H)}_{\text{scale-side supervision}}.$$

The first teaches EAA how a transition-time prediction differs from the endpoint of its own low-resolution trajectory. The second teaches CRLU how the target generator represents the same video content at two spatial scales. Separating these objectives prevents trajectory-dependent errors from contaminating the cross-resolution mapping and avoids joint backpropagation through the large frozen denoiser and the upsampler.

Endpoint Alignment Adapter

Handoff-local endpoint supervision. For each training condition, we run the frozen model with a fixed text prompt, random seed, initial noise, scheduler, and Classifier-Free Guidance (CFG) configuration. We cache the low-resolution state x_s^L immediately before the designated transition and continue the same native low-resolution trajectory to obtain its converged endpoint z_T^L. The pair (x_s^L, z_T^L) preserves the exact semantic and stochastic conditions of the transition while isolating the residual error that would otherwise be passed directly to CRLU.

Localized low-rank adaptation. EAA is implemented with LoRA (Hu et al. 2022) and is activated only at the designated transition step. The base Wan parameters remain frozen. For a linear layer W, the adapted weight is

$$W' = W + \frac{\alpha}{r} BA,$$

where r is the LoRA rank. Low-rank updates are injected into the q, k, v, and o attention projections and the ffn.0 and ffn.2 linear layers. Under the flow-matching parameterization, EAA produces the calibrated clean-latent estimate

$$\tilde{z}_s^L = x_s^L - \sigma_s v_{\phi+\Delta\theta_s}(x_s^L, s, c),$$

where σ_s is the scheduler coefficient at the transition step. We optimize

$$\mathcal{L}_{\text{EAA}} = \|\tilde{z}_s^L - z_T^L\|_1 + 0.1\|\tilde{z}_s^L - z_T^L\|_2^2.$$

For earlier transitions, where frame-wise residual errors are more likely to propagate through the scale conversion, we additionally apply a temporal residual alignment term with weight 0.05. Although the same target can be expressed as a velocity label $v^* = (x_s^L - z_T^L)/\sigma_s$, all reported models are optimized in the clean-latent domain. EAA performs no spatial upsampling; its sole role is to move the transition state toward the native low-resolution endpoint before CRLU is applied.

Cached-prefix consistency. Because EAA is inactive before step s, its low-rank parameters satisfy $\Delta\theta_k = 0$ for all $k < s$. Under deterministic sampling and matched text embeddings, initial noise, scheduler, CFG scale, and other conditioning inputs, the adapted and frozen processes therefore use identical transition functions throughout the prefix. By induction,

$$x_k^{\text{adapted}} = x_k^{\text{base}}, \qquad \forall k \leq s.$$

Consequently, a prefix state cached from the frozen model is exactly the state consumed by EAA. This property supports

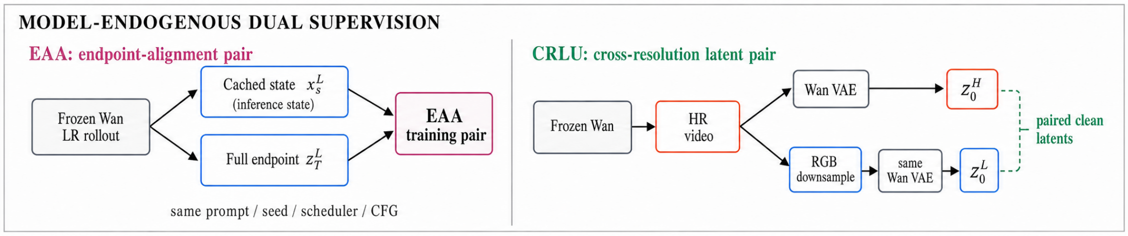

Figure 2: Overview of TrajScale. Top: a low-resolution prefix is converted into a high-resolution suffix through endpoint alignment, clean-latent upsampling, and target-resolution re-noising. Bottom: both trainable modules receive model-endogenous supervision from the same frozen Wan generator. EAA pairs a cached transition state with the endpoint of the corresponding native low-resolution trajectory, while CRLU pairs clean latents encoded from a generated high-resolution video and its RGB-downsampled counterpart.

offline pair construction and controlled comparisons among multiple transition operators without introducing prefix rollout mismatch; the same cached state can also be reused whenever an identical prompt–seed prefix is evaluated repeatedly. The equality requires transition-local activation, deterministic prefix sampling, and matched schedulers, guidance, and conditioning. Extending the adaptation to earlier or multiple steps would change subsequent states and would require on-policy student rollouts or synchronized teacher trajectories.

Cross-Resolution Latent Upsampler

Generator-endogenous scale supervision. We first use the frozen Wan model to generate high-resolution reference videos from predefined prompts and random seeds. Each video is spatially downsampled in RGB space, after which the high- and low-resolution versions are encoded independently by the same frozen Wan VAE. This produces aligned clean-latent pairs (z_0^L, z_0^H) with identical frame counts, semantics, and motion trajectories. Performing the resolution change before VAE encoding is important: CRLU learns the actual cross-resolution correspondence induced jointly by the target generator, RGB degradation, and VAE, rather than imitating an arbitrary interpolation rule inside latent space.

Spatiotemporal latent upsampling. CRLU maps a 16-channel Wan latent to a target-resolution latent while preserving the temporal dimension:

$$U_\psi^{\mathrm{CRLU}} : \mathbb{R}^{B\times16\times F\times h\times w} \to \mathbb{R}^{B\times16\times F\times H\times W}.$$

Inspired by the latent upsampler in LTX-2 (HaCohen et al. 2026), the network follows an encoder–resampler–decoder design. A 3D convolution projects the input into a 256-dimensional hidden representation, followed by four spatiotemporal residual blocks. A spatial resampler changes only the height and width, after which four additional 3D residual blocks and a final 3D convolution reconstruct the 16-channel output. The 3D blocks allow adjacent latent frames to contribute to the reconstruction without changing the total number of frames.

We instantiate the spatial resampler according to the required latent-grid ratio. For the $480\times832 \to 720\times1248$ transition, the latent grid changes from 60×104 to 90×156. CRLU first expands the channel dimension by a factor of nine, applies a $3\times$ spatial PixelShuffle, and then performs stride-2 downsampling with a fixed binomial blur kernel:

$$60\times104 \xrightarrow{\times3} 180\times312 \xrightarrow{/2} 90\times156.$$

For $368\times640 \to 720\times1248$, the latent grid changes from 46×80 to 90×156. A $2\times$ PixelShuffle first produces a 92×160 grid, which is center-cropped to the exact target size. These resolution-specific operators implement the desired rational scale changes while the surrounding spatiotemporal network learns the latent correspondence.

Cross-resolution objectives. Given $\hat{z}_0^H = U_\psi^{\mathrm{CRLU}}(z_0^L)$, CRLU is trained with

$$\mathcal{L}_{\mathrm{CRLU}} = \mathcal{L}_{\mathrm{rec}} + \lambda_{\mathrm{content}}\mathcal{L}_{\mathrm{content}} + \lambda_{\mathrm{temp}}\mathcal{L}_{\mathrm{temp}},$$

where

$$\mathcal{L}_{\mathrm{rec}} = \mathbb{E}\left[\sqrt{(\hat{z}_0^H - z_0^H)^2 + \epsilon^2}\right],$$

and

$$\mathcal{L}_{\mathrm{content}} = \|D(\hat{z}_0^H) - z_0^L\|_1, \tag{1}$$

$$\mathcal{L}_{\mathrm{temp}} = \|\Delta_F\hat{z}_0^H - \Delta_F z_0^H\|_1. \tag{2}$$

Here, D maps the prediction back to the low-resolution grid, and Δ_F denotes first-order temporal differences between adjacent latent frames. We use $\lambda_{\mathrm{content}} = 0.2$, $\lambda_{\mathrm{temp}} = 0.1$, and $\epsilon = 10^{-3}$. The three terms respectively constrain target-latent reconstruction, preservation of low-frequency content, and temporal evolution.

A low-resolution latent does not uniquely specify every high-frequency texture in its high-resolution counterpart. CRLU is therefore not tasked with deterministically hallucinating all missing detail. Instead, it produces a structurally faithful and temporally stable clean latent on the target grid. The remaining underdetermined frequencies are delegated to the frozen high-resolution suffix, preserving a clear division of labor between learned scale conversion and generative refinement.

3.3 Schedule-Consistent High-Resolution Continuation

After endpoint calibration and cross-resolution upsampling, TrajScale has a target-grid clean latent

$$\widehat{z}_s^H = U_\psi^{\mathrm{CRLU}}(\widetilde{z}_s^L),$$

but this clean estimate is not itself a valid intermediate state for the next denoising evaluation. TRR reconstructs the state at the intended scheduler level using the next-step coefficient σ_{s+1}:

$$x_{s+1}^H = (1 - \sigma_{s+1})\widehat{z}_s^H + \sigma_{s+1}\epsilon^H, \qquad \epsilon^H \sim \mathcal{N}(0, I).$$

The noise is sampled directly on the target-resolution grid. Unlike an Interpolated-Flow Transition that resizes a low-resolution noisy state or velocity field, TRR first completes the transition in the clean-latent domain and then reconstructs a high-resolution state consistent with the existing scheduler. It therefore avoids carrying a resolution-dependent noise pattern across scales and supplies the frozen denoiser with the state type it was originally trained to process.

Starting from x_{s+1}^H, the remaining evaluations are performed by the unmodified Wan denoiser in the high-resolution latent space. This suffix refines the transferred structure and uses the base model's native prior to synthesize fine textures and other spatial frequencies that are not recoverable from z_0^L alone. TrajScale consequently assigns coarse semantic layout and motion formation to the inexpensive low-resolution prefix, cross-scale structure transfer to CRLU, and high-frequency completion to a short high-resolution suffix.

The same transition rule applies to both conventional many-step samplers and temporally distilled few-step samplers. Given any valid transition index s, the sampler executes its prefix at low resolution, invokes EAA, CRLU, and TRR once, and completes the remaining scheduler states at the target resolution. The framework does not alter the base scheduler or require retraining the frozen denoiser. Concrete transition indices, scheduler settings, and quality- or efficiency-oriented operating points are specified in Section 4, since they are instantiations of the general mixed-resolution procedure rather than part of its definition.

4 Experiments

4.1 Experimental Setup

We evaluate TrajScale at both system and module levels, covering end-to-end quality–latency trade-offs, the individual and interactive effects of CRLU and EAA, and compatibility with a 4-step distilled sampler.

For the 50-step pipeline, we generate approximately 1,000 reference videos using Wan2.1 T2V-1.3B. Each video contains 81 frames at 720×1248 resolution and 16 fps. Bicubic RGB downsampling produces corresponding 480×832 and 368×640 videos, which are re-encoded by the frozen Wan VAE to construct aligned cross-resolution latent pairs. EAA training pairs consist of the low-resolution state immediately preceding a designated transition and the endpoint of the corresponding complete low-resolution trajectory.

The 4-step experiments use approximately 5,000 videos generated by Wan2.1 T2V-14B StepDistill-CfgDistill. CRLU is trained on $368 \times 640 \rightarrow 720 \times 1248$ clean latent pairs, while EAA pairs the state preceding step 3 with the complete step-4 low-resolution endpoint. Training, validation, and test partitions are separated by both text prompts and random seeds.

CRLU is optimized with AdamW for up to 50k steps using a learning rate of 1×10^{-4}, $bf16$ precision, and an EMA decay of 0.9999. EAA is trained for up to 10k steps with AdamW, a learning rate of 5×10^{-5}, and LoRA rank/scale $32/32$. Training uses four NVIDIA H100 GPUs and requires approximately 8 hours for CRLU and 33 hours for EAA. All efficiency measurements are averaged over five independent cold-start runs. Since all supervision is generated by the frozen target model, TrajScale requires neither external paired videos nor an additional upscaling teacher.

4.2 Baselines and Evaluation Protocol

System-level comparisons include **Native-HR Sampling**, which executes all denoising steps at 720p; **Trilinear Handoff**, which performs mixed-resolution inference using fixed trilinear latent interpolation; and **Endpoint Renoising**, which completes the low-resolution trajectory, applies CRLU, and performs one low-intensity high-resolution refinement step. We additionally report **Native-LR Sampling** as a low-resolution reference.

Module interactions are evaluated through a 2×2 factorial design that crosses handoff states $\{\text{Unaligned}, \text{EAA-aligned}\}$ with resolution operators $\{\text{Trilinear}, \text{CRLU}\}$. The four resulting configurations are Trilinear Handoff, EAA+Trilinear, CRLU-only, and the complete TrajScale framework. Oracle-Aligned CRLU and Joint Trajectory–Scale Mapper (JTSM) are used as diagnostic references; extended results are provided in the Supplementary Material.

For endpoint alignment, we report endpoint latent L_1, decoded PSNR and LPIPS, temporal latent differences, and sample-wise win rates. CRLU is evaluated using latent L_1, PSNR, SSIM, LPIPS, and temporal L_1. End-to-end quality is measured with VBench (Huang et al. 2024) and human evaluation. We define **VBench-5** as the unweighted mean of subject consistency, background consistency, motion smoothness, aesthetic quality, and imaging quality.

All paired configurations share identical prompts, random seeds, initial noise, schedulers, and guidance parameters. Prompt-level score differences are analyzed using 10,000 paired bootstrap iterations for 95% confidence intervals and two-sided exact sign tests. Human evaluation uses randomized, double-blind comparisons over 10 matched prompts, with three independent judges per prompt and prompt-level majority voting. Inter-rater agreement is reported in the Supplementary Material.

Efficiency measurements include end-to-end cold-start latency, peak GPU memory, low- and high-resolution denoising evaluations, and speedup over Native-HR Sampling. All EAA, CRLU, TRR, VAE, and dynamic model-loading overheads are included.

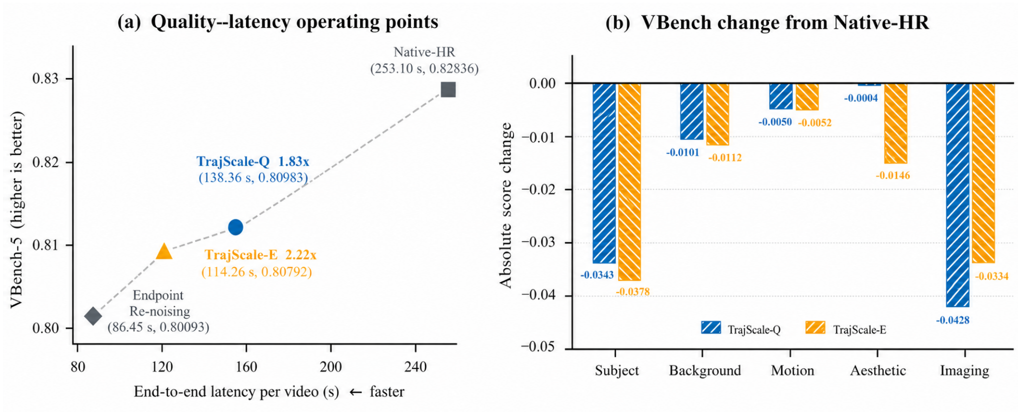

Figure 3: Quality–efficiency trade-offs relative to Native-HR Sampling. (a) TrajScale-Q and TrajScale-E provide quality- and efficiency-oriented operating points, while Endpoint Re-noising achieves higher speed at a larger quality cost. (b) Absolute per-dimension VBench changes relative to Native-HR.

Method	LR/HR evals.	Time (s) ↓	Latency red.	Speedup ↑	VBench-5 ↑	Retention ↑
Native-HR Sampling	0/50	253.10 ± 1.53	—	1.00×	**0.82836**	100.00%
TrajScale-Q (Step 40)	40/10	138.36 ± 0.50	45.33%	1.83×	0.80983	97.76%
TrajScale-E (Step 45)	45/5	$\mathbf{114.26 \pm 1.52}$	**54.85%**	**2.22×**	0.80792	97.53%

Table 1: System-level comparison against Native-HR Sampling. Timings average five cold-start runs, and quality is evaluated on ten matched prompts with all processing overheads included.

5 Results and Analysis

5.1 System-Level Quality–Efficiency Trade-off

Table 1 compares the standard 50-step Native-HR pipeline with the two principal TrajScale operating points. TrajScale-Q moves the first 40 evaluations to the low-resolution grid and retains 10 high-resolution suffix evaluations, reducing latency from 253.10 to 138.36 seconds (1.83×). TrajScale-E retains only five high-resolution evaluations and further reduces latency to 114.26 seconds (2.22×). Their VBench-5 scores are 0.80983 and 0.80792, compared with 0.82836 for Native-HR Sampling.

The Endpoint Re-noising baseline reaches 86.45 seconds (2.93×) but decreases VBench-5 to 0.80093, with inferior scores on $9/10$ prompts relative to Native-HR ($p = 0.02148$). TrajScale therefore retains a longer high-resolution suffix when additional structural and texture refinement is beneficial, while later transitions prioritize inference speed. Peak memory remains approximately 26 GiB across Native-HR, TrajScale-Q, TrajScale-E, and Endpoint Re-noising, indicating that the primary system benefit of TrajScale is reduced runtime rather than memory consumption.

5.2 Component Analysis

Cross-resolution latent upsampling. Table 2 evaluates CRLU against trilinear interpolation over 50 paired clean latent samples. For 480p → 720p, CRLU reduces latent L_1 from 0.1929 to 0.0991, corresponding to a 48.6% reduction. It also improves decoded PSNR by 5.71 dB, reduces LPIPS from 0.2358 to 0.0629, and lowers temporal L_1 by 30.8%. CRLU wins on all 50 samples for PSNR, LPIPS, and temporal consistency, and on 49/50 samples for SSIM.

The more aggressive 368p → 720p transition shows the same trend: latent L_1, LPIPS, and temporal L_1 are reduced by 33.4%, 51.1%, and 15.5%, respectively. These consistent latent, perceptual, and temporal improvements indicate that CRLU learns a substantially more faithful cross-resolution mapping than fixed interpolation.

Endpoint alignment. At an alignment intensity of 0.75, EAA reduces the residual low-resolution endpoint L_1 error by 25.82% at step 40 and 21.03% at step 45. Both settings improve all 10 evaluation samples, with paired confidence intervals excluding zero. EAA also increases decoded PSNR by 2.55 and 2.23 dB and reduces latent temporal difference errors by 23.44% and 21.24%, respectively. On the 4-step distilled model, EAA provides a smaller but consistent 5.03% endpoint-error reduction. All three comparisons have a 10/0 paired win rate and two-sided exact sign-test $p = 0.00195$; full confidence intervals are reported in Supplementary Table S2.

Alignment-intensity validation reveals a non-monotonic dependence on correction strength. An intermediate intensity of 0.75 provides the lowest endpoint error and the strongest downstream quality, whereas weaker or full-strength corrections are less effective. This behavior supports calibrated rather than unconstrained endpoint correction.

Factorial effects. The 2×2 analysis in Table 3 shows that CRLU provides the principal broad quality gain, improving VBench-5 by approximately 0.031–0.041 across the three evaluated pipelines. EAA serves a more targeted role. With CRLU fixed, it improves the imaging-quality dimension by +0.00599, +0.00254, and +0.00479 at step 40, step 45, and on the 4-step sampler, respectively. These detail-oriented improvements are distributed across only one component of the five-dimensional aggregate, and therefore appear as smaller changes in the averaged VBench-5 score.

Input → Output	Operator	Latent $L_1 \downarrow$	PSNR ↑	SSIM ↑	LPIPS ↓	Temp. $L_1 \downarrow$
480p → 720p	Trilinear	0.1929	24.29	0.7159	0.2358	0.02121
480p → 720p	CRLU	**0.0991**	**30.00**	**0.8696**	**0.0629**	**0.01468**
368p → 720p	Trilinear	0.2444	23.42	0.6740	0.3352	0.02388
368p → 720p	CRLU	**0.1628**	**24.10**	**0.7592**	**0.1640**	**0.02017**

Table 2: CRLU versus trilinear interpolation on paired clean latents ($n = 50$).

Sampler / transition	Trilinear	EAA+Tri.	CRLU-only	TrajScale	CRLU gain	EAA gain	Total gain
Wan50 / Step 40 (TrajScale-Q)	0.77812	0.77901	0.80896	**0.80983**	+0.03085	+0.00087	+0.03171
Wan50 / Step 45 (TrajScale-E)	0.76776	0.76734	**0.80842**	0.80792	+0.04066	-0.00050	+0.04016
Distill4 / Step 3-of-4	0.81796	0.81909	0.85670	**0.85680**	+0.03874	+0.00010	+0.03884

Table 3: End-to-end factorial results on VBench-5 ($n = 10$). "EAA gain" denotes the marginal effect of enabling EAA with CRLU fixed.

Comparison	Details	Overall	Temporal
EAA vs. Unaligned, CRLU (S40)	8/2/0	6/2/2	1/9/0
EAA vs. Unaligned, CRLU (S45)	**9/0/1**	**6/0/4**	0/8/2
CRLU-only vs. Trilinear (S45)	**10/0/0**	**10/0/0**	**8/0/2**
TrajScale-E vs. Trilinear (S45)	**10/0/0**	**10/0/0**	**9/0/1**

Table 4: Prompt-level majority votes (W/L/T) over ten prompts with three raters per prompt. Bold entries have exact sign-test $p < 0.05$.

The results support a clear division of labor. CRLU restores the cross-resolution latent representation and supplies the dominant improvements in motion smoothness, aesthetics, and overall imaging quality. EAA further calibrates the handoff state toward the converged low-resolution endpoint, enriching localized textures and fine structures that may be underrepresented by aggregate automatic metrics.

5.3 Perceptual Effects and Distilled-Sampler Compatibility

Human evaluation confirms that the detail recovery introduced by EAA is perceptually visible. With the same CRLU operator, EAA receives fine-detail win/loss/tie counts of $8/2/0$ at step 40 and $9/0/1$ at step 45. Overall-quality judgments are also positive, reaching $6/2/2$ and $6/0/4$, respectively. These preferences support the dimension-level VBench observation that EAA selectively improves imaging quality and localized structure.

The additional fine-scale content introduces a spatial–temporal trade-off: temporal stability favors the unaligned CRLU state in both settings. This suggests that EAA shifts the operating point toward sharper local rendering, while CRLU and the remaining high-resolution suffix provide the broader spatial and temporal reconstruction.

CRLU provides the broader perceptual improvement: at step 45, both CRLU-only and TrajScale-E obtain $10/10$ prompt-level wins over trilinear handoff for fine details and overall quality, while also improving temporal stability. EAA adds a narrower refinement of local detail on top of this cross-resolution reconstruction.

TrajScale is also compatible with the 4-step distilled Wan sampler. On the 368p trajectory, Trilinear Handoff achieves a VBench-5 score of 0.81796, whereas TrajScale-D4 reaches 0.85680. The associated cold-start latency increases only from 167.09 to 170.29 seconds. As in the 50-step setting, CRLU supplies the primary cross-resolution quality improvement, while EAA performs localized endpoint calibration and improves the imaging-quality component.

Qualitative comparisons in Figure 1(a) further illustrate CRLU's recovery of high-frequency textures and EAA's localized adjustments. Extended spatial crops and temporal sequences are provided in Figure S1. The content-matched `Native-HR (estimated)` reference is produced from a complete 368p trajectory and is used only for frame-level structural comparison; true 720p Native-HR Sampling is used for the quantitative evaluation in Table 1.

6 Discussion and Limitations

TrajScale assigns endpoint calibration, cross-resolution clean-latent upsampling, and schedule-consistent state reconstruction to EAA, CRLU, and TRR. This decomposition keeps CRLU in a stable clean-latent domain while the frozen high-resolution suffix recovers underdetermined detail. Endpoint metrics, imaging-quality gains, and human preferences indicate that EAA improves localized structure and perceptual detail, although its effect depends on the CRLU input distribution and the remaining suffix length.

EAA and CRLU use model-endogenous self-distillation: the frozen Wan model supplies high-resolution references for CRLU and complete low-resolution trajectories for EAA, avoiding external paired data, standalone upscalers, and additional teachers.

Several limitations remain. CRLU is tied to the target VAE, generator distribution, and scaling configuration, so cross-model transfer is unverified. A distribution gap persists between its training inputs and native low-resolution trajectory states. Fixed handoff steps also require step-specific EAA parameters, motivating timestep-conditioned adapters for adaptive handoffs. Finally, TrajScale addresses in-trajectory spatial scaling rather than real-world degradations such as compression, blur, or sensor noise. It inherits the base generator's content risks and safety requirements.

7 Conclusion

We presented TrajScale, a decoupled mixed-resolution sampling framework that combines a low-resolution prefix with a short high-resolution suffix. EAA calibrates the handoff prediction toward the converged low-resolution endpoint, CRLU upsamples clean latents across spatial scales, and TRR reconstructs a schedule-consistent high-resolution state. All trainable components use supervision generated by the frozen base model.

TrajScale-Q and TrajScale-E achieve $1.83\times$ and $2.22\times$ speedups while retaining 97.76% and 97.53% of Native-HR VBench-5 performance. CRLU consistently outperforms trilinear interpolation, while EAA improves endpoint alignment and localized visual detail across standard and distilled samplers.

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